BASE
DE
DATOS

NEREA MONTOYA OLIVARES

EXERCISE 1

a) Create the object type *Member* with the following attributes:

idMember CHAR(4),
fullname VARCHAR2(100),
username VARCHAR2(15),
phone VARCHAR2(15),
email VARCHAR2(100),
dateEntry DATE

```
CREATE OR REPLACE TYPE Member AS OBJECT (
        idMember CHAR(4),
        fullName VARCHAR2(100),
        userName VARCHAR2(15),
        phone VARCHAR2(15),
        email VARCHAR2(100),
        dateEntry DATE
) NOT FINAL;
/
```

b) Create, the object type *Teacher* as an inherited type of *Member* with the following attributes:

title VARCHAR2(100),
salary NUMBER(8,2),
category VARCHAR2(20)

```
CREATE OR REPLACE TYPE Teacher UNDER Member (
        title VARCHAR2(100),
        salary NUMBER(8,2),
        category VARCHAR2(20)
 );
/
```

c) Create, the object type *Groups* with the following attributes:

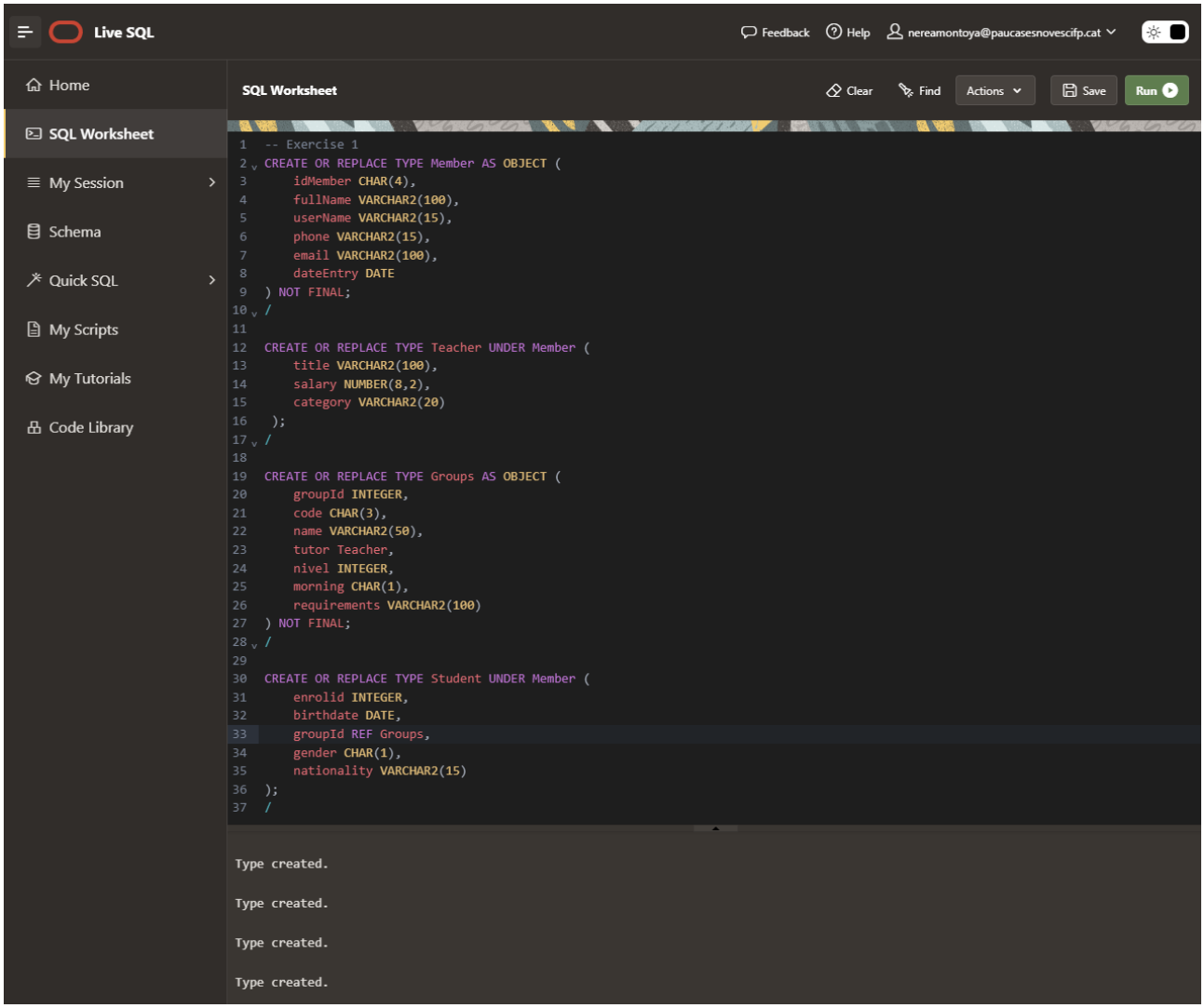
groupId INTEGER,
code CHAR(3),
name VARCHAR2(50),
tutor Teacher,
level INTEGER,
morning CHAR(1),
requirements VARCHAR(100)

```
CREATE OR REPLACE TYPE Groups AS OBJECT (
        groupId INTEGER,
        code CHAR(3),
        name VARCHAR2(50),
        tutor Teacher,
        nivel INTEGER,
        morning CHAR(1),
        requirements VARCHAR2(100)
) NOT FINAL;
/
```

d) Create the object type *Student* as an inherited type of *Member* with the following attributes:

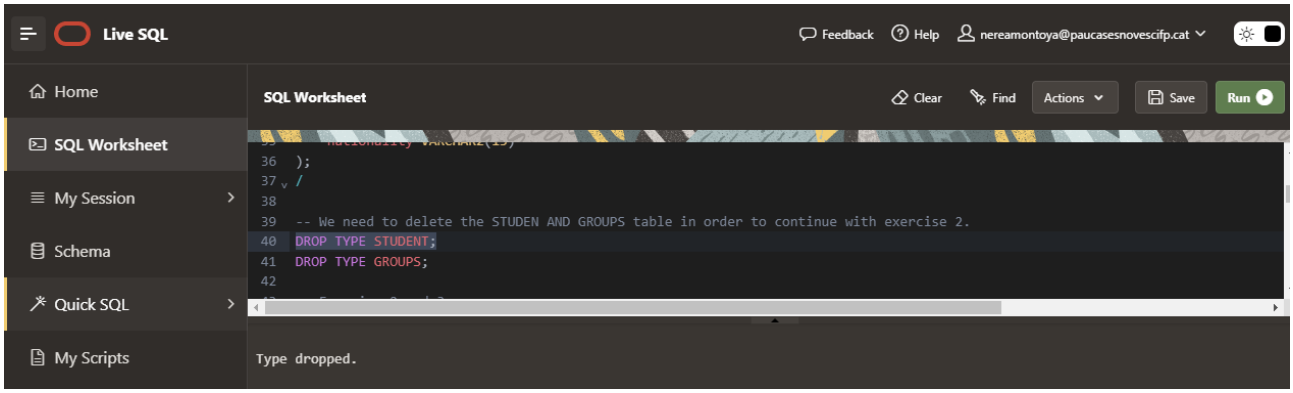
enrolid INTEGER,
birthdate DATE,
groupId REF Groups,
gender CHAR(1),
nationality VARCHAR2(15)

```
CREATE OR REPLACE TYPE Student UNDER Member (
        enrolid INTEGER,
        birthdate DATE,
        groupId REF Groups,
        gender CHAR(1),
        nationality VARCHAR2(15)
);
/
```

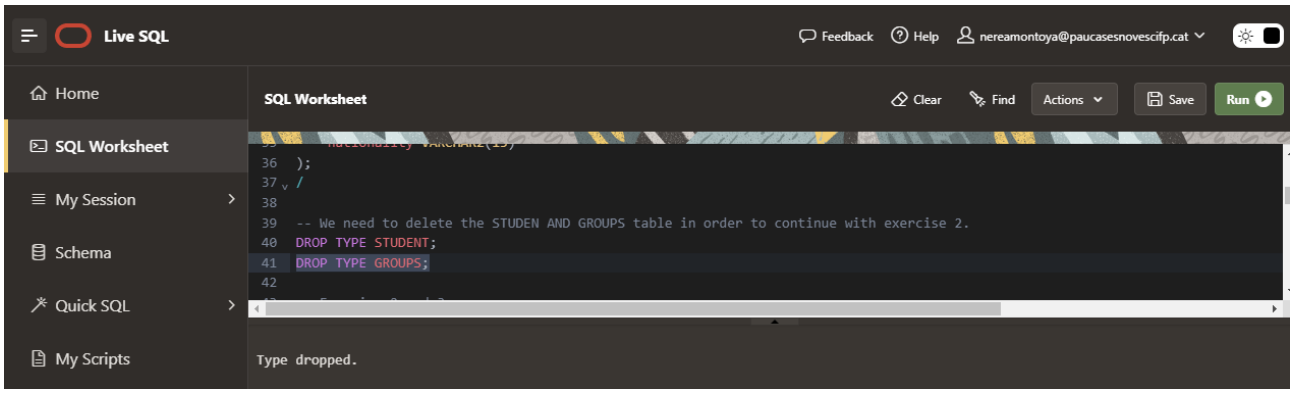


- Once you have created all the types, for keep doing the exercises 2 and 3 for taking only the type teacher you have to drop the types STUDENT and GROUPS:

DROP TYPE STUDENT



DROP TYPE GROUPS



EXERCISE 2

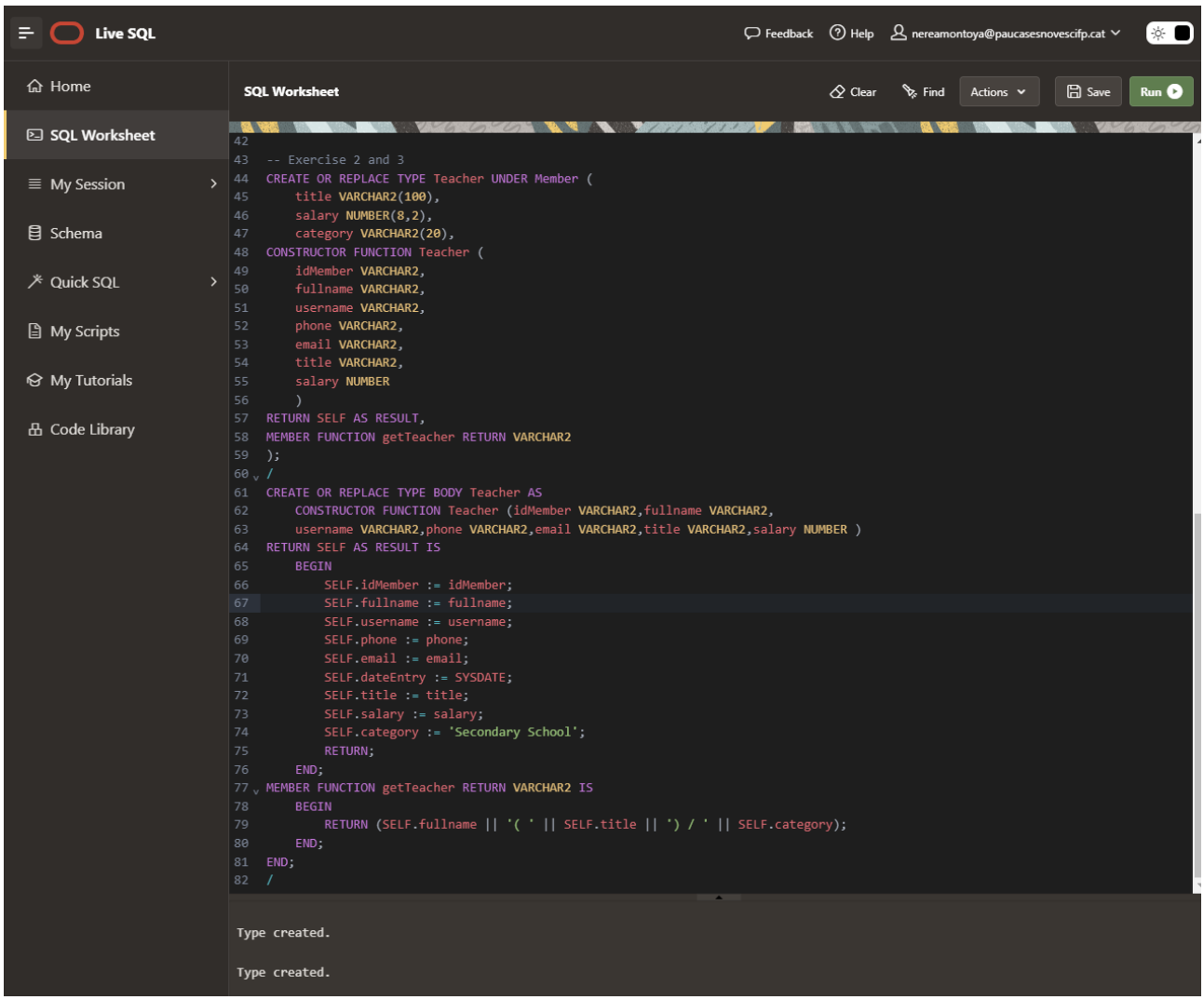
Create a constructor method for the object type Teacher with the following parameters:

- idMember,
- fullname,
- username,
- phone,
- email
- title
- salary

The method must:

- assign a default value of Secondary School for category and current date for dateEntry

```
-- Exercise 2 and 3
CREATE OR REPLACE TYPE Teacher UNDER Member (
        title VARCHAR2(100),
        salary NUMBER(8,2),
        category VARCHAR2(20),
CONSTRUCTOR FUNCTION Teacher (
        idMember VARCHAR2,
        fullname VARCHAR2,
        username VARCHAR2,
        phone VARCHAR2,
        email VARCHAR2,
        title VARCHAR2,
        salary NUMBER
        )
RETURN SELF AS RESULT,
MEMBER FUNCTION getTeacher RETURN VARCHAR2
);
/
CREATE OR REPLACE TYPE BODY Teacher AS
        CONSTRUCTOR FUNCTION Teacher (idMember VARCHAR2,fullname VARCHAR2,
        username VARCHAR2,phone VARCHAR2,email VARCHAR2,title VARCHAR2,salary NUMBER )
RETURN SELF AS RESULT IS
        BEGIN
                SELF.idMember := idMember;
                SELF.fullname := fullname;
                SELF.username := username;
                SELF.phone := phone;
                SELF.email := email;
                SELF.dateEntry := SYSDATE;
                SELF.title := title;
                SELF.salary := salary;
                SELF.category := 'Secondary School';
                RETURN;
        END;
MEMBER FUNCTION getTeacher RETURN VARCHAR2 IS
        BEGIN
                RETURN (SELF.fullname || '( ' || SELF.title || ') / ' || SELF.category);
        END;
END;
/
```

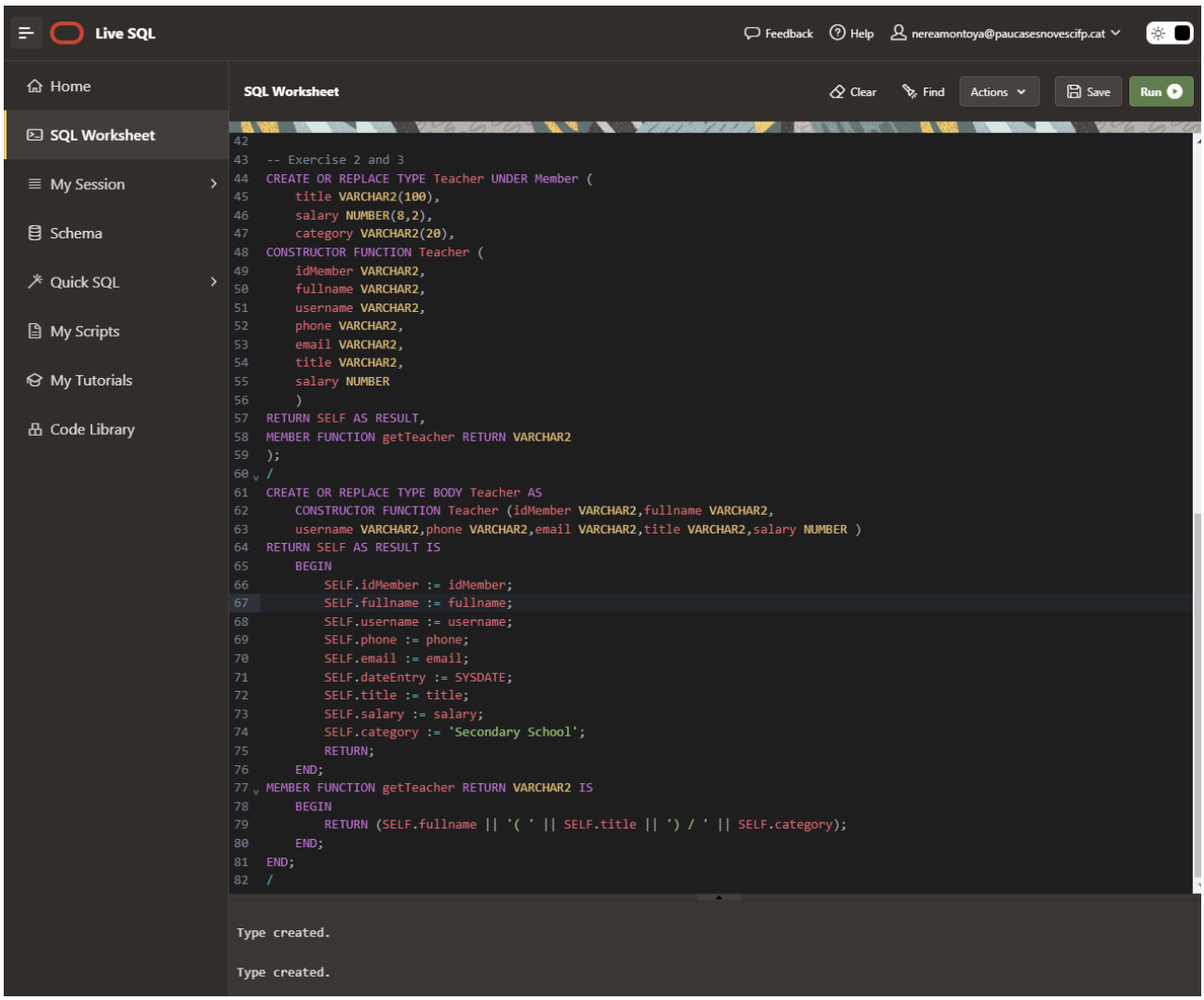

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43 -- Exercise 2 and 3
44 CREATE OR REPLACE TYPE Teacher UNDER Member (
45 title VARCHAR2(100),
46 salary NUMBER(8,2),
47 category VARCHAR2(20),
48 CONSTRUCTOR FUNCTION Teacher (
49 idMember VARCHAR2,
50 fullname VARCHAR2,
51 username VARCHAR2,
52 phone VARCHAR2,
53 email VARCHAR2,
54 title VARCHAR2,
55 salary NUMBER
56)
57 RETURN SELF AS RESULT,
58 MEMBER FUNCTION getTeacher RETURN VARCHAR2
59);
60 /
61 CREATE OR REPLACE TYPE BODY Teacher AS
62 CONSTRUCTOR FUNCTION Teacher (idMember VARCHAR2,fullname VARCHAR2,
63 username VARCHAR2,phone VARCHAR2,email VARCHAR2,title VARCHAR2,salary NUMBER)
64 RETURN SELF AS RESULT IS
65 BEGIN
66 SELF.idMember := idMember;
67 SELF.fullname := fullname;
68 SELF.username := username;
69 SELF.phone := phone;
70 SELF.email := email;
71 SELF.dateEntry := SYSDATE;
72 SELF.title := title;
73 SELF.salary := salary;
74 SELF.category := 'Secondary School';
75 RETURN;
76 END;
77 MEMBER FUNCTION getTeacher RETURN VARCHAR2 IS
78 BEGIN
79 RETURN (SELF.fullname || '(' || SELF.title || ') / ' || SELF.category);
80 END;
81 END;
82 /
Type created.
Type created.

EXERCISE 3

Create a "getTeacher" method for the *Teacher* object type that allows you to obtain the member data with the following format:
"fullname (title) / category"

For example, for a ***Teacher*** the method should show:
Juliette Smith (Bachelor in Computer Science) / Secondary School.

```
-- Exercise 2 and 3
CREATE OR REPLACE TYPE Teacher UNDER Member (
        title VARCHAR2(100),
        salary NUMBER(8,2),
        category VARCHAR2(20),
CONSTRUCTOR FUNCTION Teacher (
        idMember VARCHAR2,
        fullname VARCHAR2,
        username VARCHAR2,
        phone VARCHAR2,
        email VARCHAR2,
        title VARCHAR2,
        salary NUMBER
        )
RETURN SELF AS RESULT,
MEMBER FUNCTION getTeacher RETURN VARCHAR2
);
/
CREATE OR REPLACE TYPE BODY Teacher AS
        CONSTRUCTOR FUNCTION Teacher (idMember VARCHAR2,fullname VARCHAR2,
        username VARCHAR2,phone VARCHAR2,email VARCHAR2,title VARCHAR2,salary NUMBER )
RETURN SELF AS RESULT IS
        BEGIN
                SELF.idMember := idMember;
                SELF.fullname := fullname;
                SELF.username := username;
                SELF.phone := phone;
                SELF.email := email;
                SELF.dateEntry := SYSDATE;
                SELF.title := title;
                SELF.salary := salary;
                SELF.category := 'Secondary School';
                RETURN;
        END;
MEMBER FUNCTION getTeacher RETURN VARCHAR2 IS
        BEGIN
                RETURN (SELF.fullname || '( ' || SELF.title || ') / ' || SELF.category);
        END;
END;
/
```


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43 -- Exercise 2 and 3
44 CREATE OR REPLACE TYPE Teacher UNDER Member (
45 title VARCHAR2(100),
46 salary NUMBER(8,2),
47 category VARCHAR2(20),
48 CONSTRUCTOR FUNCTION Teacher (
49 idMember VARCHAR2,
50 fullname VARCHAR2,
51 username VARCHAR2,
52 phone VARCHAR2,
53 email VARCHAR2,
54 title VARCHAR2,
55 salary NUMBER
56)
57 RETURN SELF AS RESULT,
58 MEMBER FUNCTION getTeacher RETURN VARCHAR2
59);
60 /
61 CREATE OR REPLACE TYPE BODY Teacher AS
62 CONSTRUCTOR FUNCTION Teacher (idMember VARCHAR2,fullname VARCHAR2,
63 username VARCHAR2,phone VARCHAR2,email VARCHAR2,title VARCHAR2,salary NUMBER)
64 RETURN SELF AS RESULT IS
65 BEGIN
66 SELF.idMember := idMember;
67 SELF.fullname := fullname;
68 SELF.username := username;
69 SELF.phone := phone;
70 SELF.email := email;
71 SELF.dateEntry := SYSDATE;
72 SELF.title := title;
73 SELF.salary := salary;
74 SELF.category := 'Secondary School';
75 RETURN;
76 END;
77 MEMBER FUNCTION getTeacher RETURN VARCHAR2 IS
78 BEGIN
79 RETURN (SELF.fullname || '(' || SELF.title || ') / ' || SELF.category);
80 END;
81 END;
82 /
Type created.
Type created.

EXERCISE 4

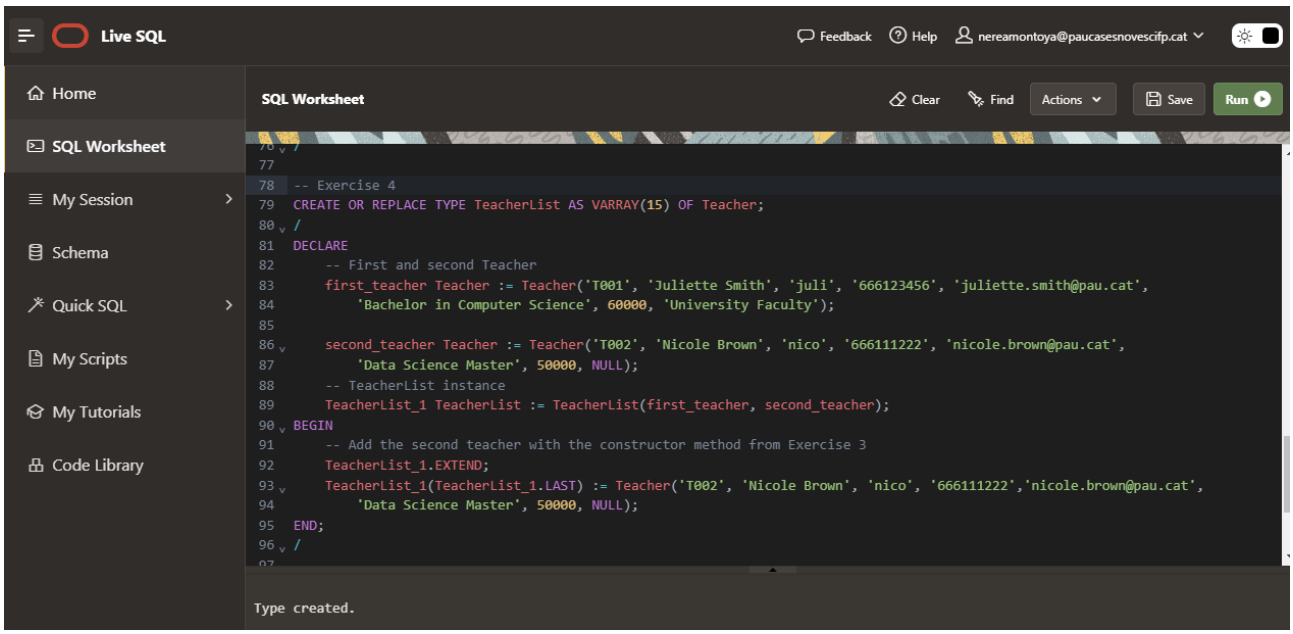
Create a VARRAY collection called TeacherList to store up to 15 Teacher objects. Later, create a TeacherList instance called TeacherList_1 with the data of the following two teachers:

a) The first teacher to insert in the list is:	b) The second teacher (using the constructor method done in the exercise 3) to
idMember T001, fullname Juliette Smith, username juli, phone 666123456, email juliette.smith@pau.cat dateEntry 01/04/2023 title Bachelor in Computer Science salary 60000, category University Faculty	insert in the list is: idMember T002 fullname Nicole Brown username nico phone 666111222 email nicole.brown@pau.cat title Data Science Master salary 50000

```
CREATE OR REPLACE TYPE TeacherList AS VARRAY(15) OF Teacher;
DECLARE
   -- First and Second Teacher
   first_teacher Teacher := Teacher('T001', 'Juliette Smith', 'juli', '666123456', 'juliette.smith@pau.cat',
         'Bachelor in Computer Science', 60000, 'University Faculty');

   second_teacher Teacher := Teacher('T002', 'Nicole Brown', 'nico', '666111222',
         'nicole.brown@pau.cat', 'Data Science Master', 50000, NULL);

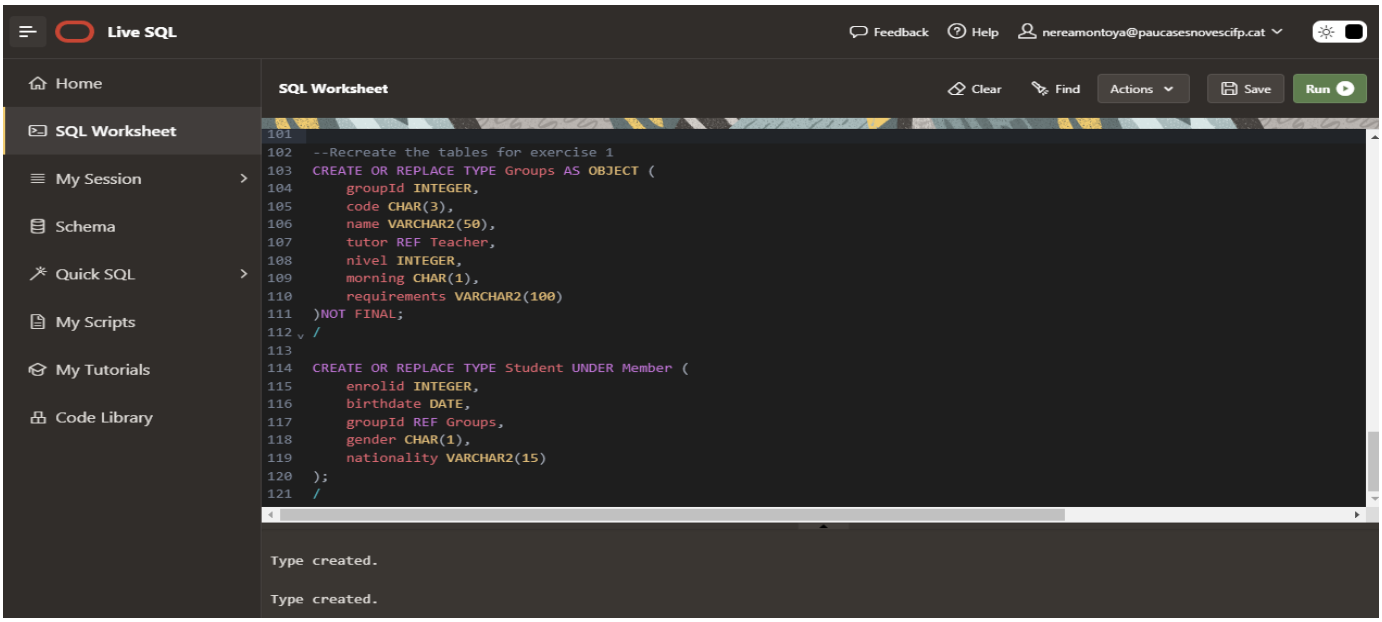
   -- TeacherList instance
   TeacherList_1 TeacherList := TeacherList(first_teacher, second_teacher);
BEGIN
   -- Add the second teacher with the constructor method from exercise 3
   TeacherList_1.EXTEND;
   TeacherList_1(TeacherList_1.LAST) := Teacher('T002', 'Nicole Brown', 'nico', '666111222',
         'nicole.brown@pau.cat', 'Data Science Master', 50000, NULL);
END;
/
```



- For the next exercises you need to create again the types STUDENT and GROUPS:

```
--Recreate the tables for exercise 1
CREATE OR REPLACE TYPE Groups AS OBJECT (
        groupId INTEGER,
        code CHAR(3),
        name VARCHAR2(50),
        tutor REF Teacher,
        nivel INTEGER,
        morning CHAR(1),
        requirements VARCHAR2(100)
)NOT FINAL;
/

CREATE OR REPLACE TYPE Student UNDER Member (
        enrolid INTEGER,
        birthdate DATE,
        groupId REF Groups,
        gender CHAR(1),
        nationality VARCHAR2(15)
);
/
```



EXERCISE 5

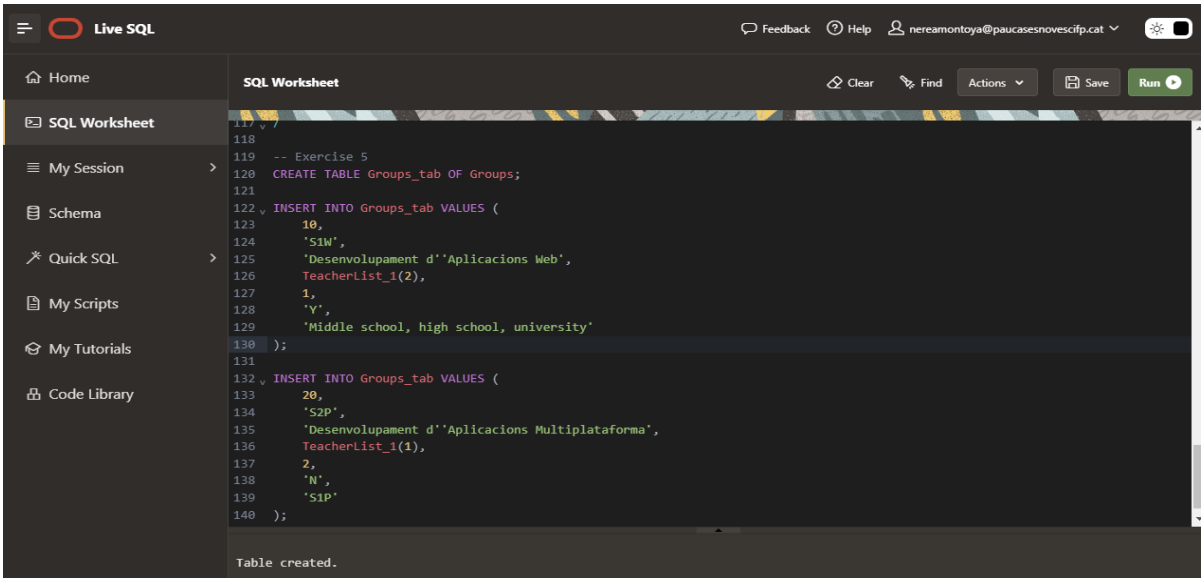
Create a Groups_tab table of Groups objects. Insert the following rows into this table:

groupId: 10 code: S1W name: Desenvolupament d'Aplicacions Web tutor: object created in the exercise 4 for the second teacher level: 1 morning: Y requirements: Middle school, high school, university	groupId: 20 code: S2P name: Desenvolupament d'Aplicacions Multiplataforma tutor: object created in the exercise 4 in the first position of the list (it must be taken of the list) level: 2 morning: N requirements: S1P

```
CREATE TABLE Groups_tab OF Groups;

INSERT INTO Groups_tab VALUES (
        10,
        'S1W',
        'Desenvolupament d''Aplicacions Web',
        TeacherList_1(2),
        1,
        'Y',
        'Middle school, high school, university'
);

INSERT INTO Groups_tab VALUES (
        20,
        'S2P',
        'Desenvolupament d''Aplicacions Multiplataforma',
        TeacherList_1(1),
        2,
        'N',
        'S1P'
);
/
```



EXERCISE 6

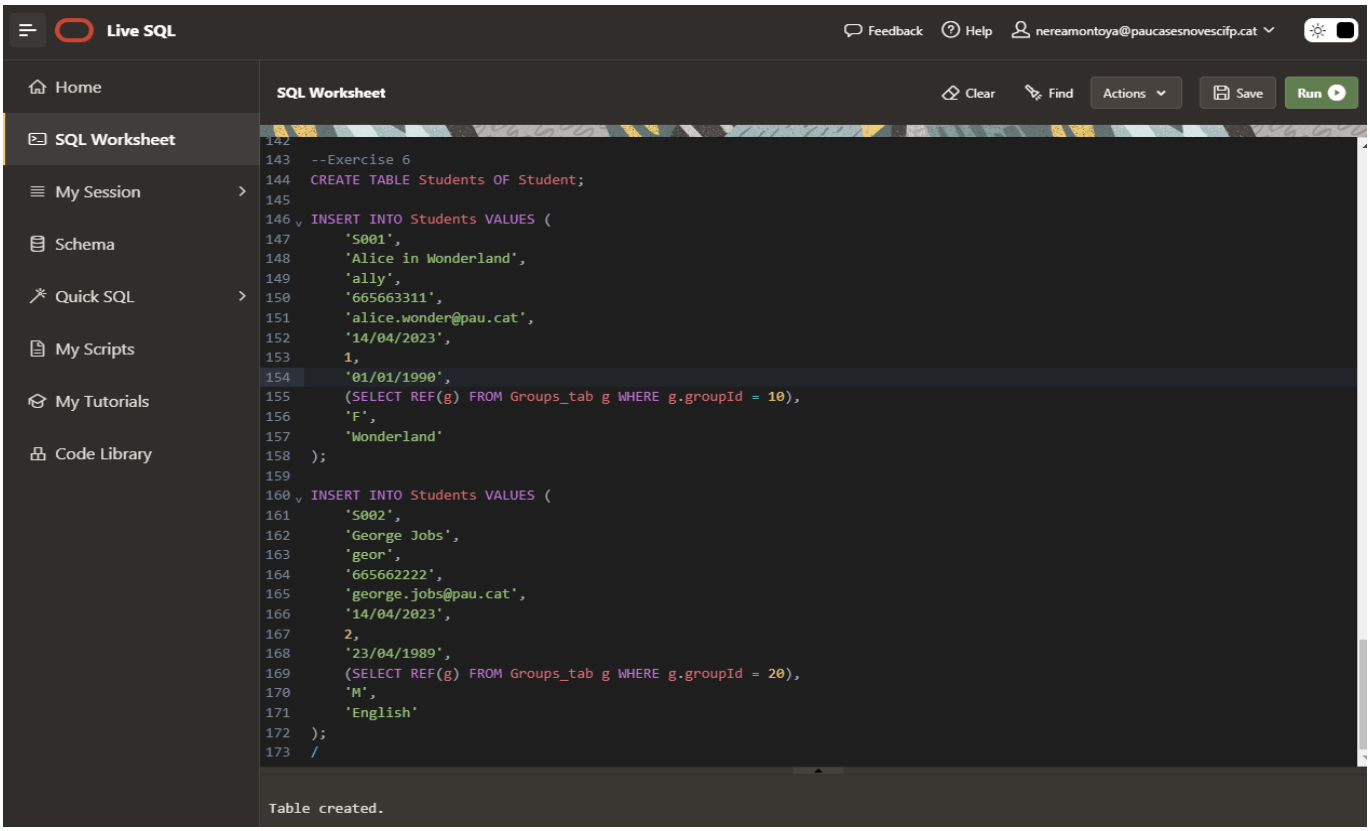
Create a Students table of Student objects. Insert two Student objects into this table.

The first one with the data:	**The second one with the data:**
idMember S001 fullname Alice in Wonderland username ally phone 665663311 email alice.wonder@pau.cat dateEntry 14/04/2023 enrolid 1 birthdate 01/01/1990 groupId Reference to the group with groupId 10 gender F nationality Wonderland	idMember S002 fullname George Jobs username geor phone 665662222 email george.jobs@pau.cat dateEntry 14/04/2023 enrolid 2 birthdate 23/04/1989 groupId Reference to the group with groupId 20 gender M nationality English

```
CREATE TABLE Students OF Student;

INSERT INTO Students VALUES (
        'S001',
        'Alice in Wonderland',
        'ally',
        '665663311',
        'alice.wonder@pau.cat',
        '14/04/2023',
        1,
        '01/01/1990',
        (SELECT REF(g) FROM Groups_tab g WHERE g.groupId = 10),
        'F',
        'Wonderland'
);

INSERT INTO Students VALUES (
        'S002',
        'George Jobs',
        'geor',
        '665662222',
        'george.jobs@pau.cat',
        '14/04/2023',
        2,
        '23/04/1989',
        (SELECT REF(g) FROM Groups_tab g WHERE g.groupId = 20),
        'M',
        'English'
);
/
```

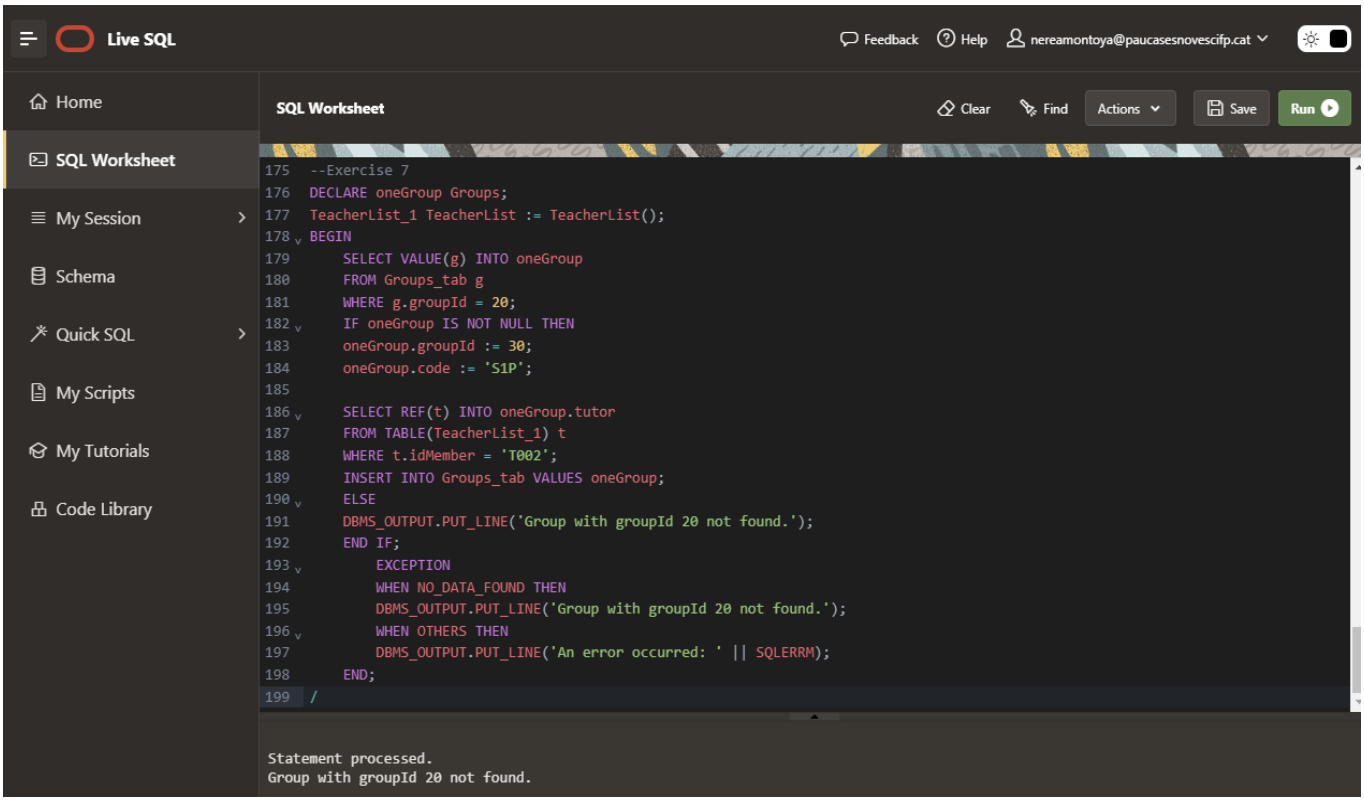

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--Exercise 6
CREATE TABLE Students OF Student;
INSERT INTO Students VALUES (
'S001',
'Alice in Wonderland',
'ally',
'665663311',
'alice.wonder@pau.cat',
'14/04/2023',
1,
'01/01/1990',
(SELECT REF(g) FROM Groups_tab g WHERE g.groupId = 10),
'F',
'Wonderland'
);
INSERT INTO Students VALUES (
'S002',
'George Jobs',
'geor',
'665662222',
'george.jobs@pau.cat',
'14/04/2023',
2,
'23/04/1989',
(SELECT REF(g) FROM Groups_tab g WHERE g.groupId = 20),
'M',
'English'
);
/
Table created.

EXERCISE 7

Obtain the group with groupId 20 from the Groups_tab table and assign it to a variable called oneGroup. Later, modify the groupId attribute saved in the variable oneGroup and assign it the value 30, also modify the code attribute and assign it the value S1P, and its tutor must be the second tutor that had been previously created in the exercise 4. Finally, insert that group in the Groups_tab table.

```
--Exercise 7
DECLARE oneGroup Groups;
TeacherList_1 TeacherList := TeacherList();
BEGIN
        SELECT VALUE(g) INTO oneGroup
        FROM Groups_tab g
        WHERE g.groupId = 20;
        IF oneGroup IS NOT NULL THEN
        oneGroup.groupId := 30;
        oneGroup.code := 'S1P';

        SELECT REF(t) INTO oneGroup.tutor
        FROM TABLE(TeacherList_1) t
        WHERE t.idMember = 'T002';
        INSERT INTO Groups_tab VALUES oneGroup;
        ELSE
        DBMS_OUTPUT.PUT_LINE('Group with groupId 20 not found.');
        END IF;
                EXCEPTION
                WHEN NO_DATA_FOUND THEN
                DBMS_OUTPUT.PUT_LINE('Group with groupId 20 not found.');
                WHEN OTHERS THEN
                DBMS_OUTPUT.PUT_LINE('An error occurred: ' || SQLERRM);
        END;
/
```


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--Exercise 7
DECLARE oneGroup Groups;
TeacherList_1 TeacherList := TeacherList();
BEGIN
SELECT VALUE(g) INTO oneGroup
FROM Groups_tab g
WHERE g.groupId = 20;
IF oneGroup IS NOT NULL THEN
oneGroup.groupId := 30;
oneGroup.code := 'S1P';
SELECT REF(t) INTO oneGroup.tutor
FROM TABLE(TeacherList_1) t
WHERE t.idMember = 'T002';
INSERT INTO Groups_tab VALUES oneGroup;
ELSE
DBMS_OUTPUT.PUT_LINE('Group with groupId 20 not found.');
END IF;
EXCEPTION
WHEN NO_DATA_FOUND THEN
DBMS_OUTPUT.PUT_LINE('Group with groupId 20 not found.');
WHEN OTHERS THEN
DBMS_OUTPUT.PUT_LINE('An error occurred: ' || SQLERRM);
END;
/
Statement processed.
Group with groupId 20 not found.